Phase 1 — Project Planning & Software Design

Objective

Create a complete software design before development begins.

Phase 1 Roadmap

```
Phase 1
│
├── Step 1.1 Project Overview
├── Step 1.2 Problem Statement
├── Step 1.3 Project Objectives
├── Step 1.4 Scope
├── Step 1.5 Stakeholders
├── Step 1.6 User Roles
├── Step 1.7 Functional Requirements
├── Step 1.8 Non-Functional Requirements
├── Step 1.9 Business Rules
├── Step 1.10 Assumptions & Constraints
├── Step 1.11 User Stories
├── Step 1.12 Use Case Diagram
├── Step 1.13 Activity Diagrams
├── Step 1.14 Sequence Diagrams
├── Step 1.15 ER Diagram (High-Level)
└── Step 1.16 Final SRS Review
```

Step 1.1 — Project Overview

Project Title

E-Commerce Platform

Project Description

The E-Commerce Platform is a full-stack web application that enables customers to browse products, manage shopping carts, place secure online orders, and track purchases. Administrators

can manage products, categories, inventory, customers, orders, coupons, and sales reports through a dedicated dashboard.

The platform aims to provide a secure, scalable, and user-friendly online shopping experience while demonstrating modern full-stack development practices using React, Node.js, Express.js, MySQL, JWT authentication, and third-party integrations such as Stripe and Cloudinary.

Project Type

- Full-Stack Web Application
- Business Management System
- Business-to-Consumer (B2C)

Target Users

Customer

- Browse products
- Purchase products
- Track orders

Administrator

- Manage the platform
- Manage inventory
- Process orders
- Monitor sales

Main Modules

Authentication

- Register
- Login
- Logout
- Forgot Password
- Email Verification

Product Management

- Products
- Categories
- Brands
- Product Variants
- Inventory

Shopping

- Cart
- Wishlist
- Checkout
- Coupons

Orders

- Place Orders
- Payment
- Order Tracking
- Order History

Reviews

- Product Reviews
- Ratings

Admin

- Dashboard
- Product Management
- Customer Management
- Order Management
- Reports
- Analytics

Technologies

Frontend

- React
- Vite
- Material UI
- React Router
- Axios
- React Hook Form
- Yup
- TanStack Query

Backend

- Node.js
- Express.js
- MySQL
- JWT
- Bcrypt
- Multer
- Cloudinary
- Nodemailer
- Stripe
- Redis

Deployment

- Vercel
- Render
- Aiven MySQL

Expected Duration

Approximately **10–14 weeks**, depending on pace and feature depth.

Expected Learning Outcomes

By completing this project, you will gain practical experience with:

- Designing a production-style full-stack application
- Building secure authentication and authorization
- Creating RESTful APIs
- Managing relational databases
- Integrating payment gateways
- Uploading and managing cloud-hosted images
- Optimizing performance with caching
- Deploying modern web applications
- Writing clean, modular, and maintainable code

Phase 1 — Step 1.2: Problem Statement

Purpose

A problem statement clearly defines **what problem the system solves**, **who experiences the problem**, and **why a new solution is needed**.

Problem Statement

The rapid growth of online shopping has increased customer expectations for fast, secure, and convenient purchasing experiences. Many small and medium-sized businesses either rely on physical stores or use outdated systems that lack essential e-commerce features such as secure online payments, real-time inventory management, efficient order processing, and personalized customer experiences.

Customers often face difficulties such as limited product search capabilities, lack of product filtering, delayed order updates, and complicated checkout processes. At the same time, business administrators struggle with manually managing products, tracking inventory, processing customer orders, and generating sales reports, which increases operational costs and the likelihood of human error.

To address these challenges, a modern E-Commerce Platform is required. The proposed system will provide customers with an intuitive shopping experience, including product search, shopping cart management, wishlist functionality, secure online payments, and order tracking. It

will also provide administrators with comprehensive tools to manage products, inventory, customers, orders, coupons, and business analytics from a centralized dashboard.

The platform will be designed using modern web technologies and best software engineering practices to ensure security, scalability, maintainability, and high performance.

Problems Identified

Customer Problems

- Difficulty finding products quickly.
- Limited search and filtering options.
- No personalized shopping experience.
- Insecure or inconvenient payment methods.
- Lack of real-time order tracking.
- Slow checkout process.

Business Problems

- Manual product management.
- Manual inventory updates.
- Difficulty tracking orders.
- Limited sales reporting.
- High risk of inventory errors.
- Time-consuming customer management.

Technical Problems

- Outdated application architecture.
- Poor scalability.
- Weak security.
- No API-based design.
- Inefficient image management.
- Slow performance under high traffic.

Proposed Solution

The proposed E-Commerce Platform will provide:

- Secure user authentication and authorization.
- Comprehensive product catalog management.
- Shopping cart and wishlist functionality.
- Secure online payments through Stripe.
- Automated inventory updates.
- Real-time order status management.
- Sales analytics and reporting.
- Cloud-based image storage.
- Responsive user interface.
- RESTful APIs for frontend and backend communication.

Benefits

For Customers

- Faster product discovery.
- Secure online shopping.
- Easy order tracking.
- Better shopping experience.
- Mobile-friendly interface.

For Administrators

- Centralized management.
- Improved inventory control.
- Faster order processing.
- Better business insights.
- Reduced manual work.

For Developers (Portfolio Value)

- Production-style architecture.
- Payment gateway integration.
- Cloud storage integration.
- Advanced authentication.
- Scalable system design.
- Real-world deployment experience.

Phase 1 — Step 1.3: Project Objectives

Purpose

Project objectives describe the **overall goal** and the **specific outcomes** the system should accomplish.

---

1. Overall Objective

To design and develop a secure, scalable, and user-friendly full-stack E-Commerce Platform that enables customers to purchase products online while providing administrators with efficient tools to manage products, inventory, orders, customers, coupons, and business analytics.

---

2. Specific Objectives

Objective 1 — User Authentication

Develop a secure authentication system with user registration, login, password recovery, email verification, and role-based access control using JWT.

Objective 2 — Product Management

Allow administrators to create, update, delete, and organize products by categories, brands, images, and variants such as size and color.

Objective 3 — Product Discovery

Provide customers with advanced product search, filtering, sorting, and pagination to quickly find relevant products.

Objective 4 — Shopping Experience

Implement a shopping cart and wishlist that allow customers to manage products before purchasing.

Objective 5 — Secure Checkout

Provide a secure checkout process with shipping details, coupon application, tax calculation, and Stripe payment integration.

Objective 6 — Order Management

Enable customers to place orders, view order history, and track order status, while allowing administrators to process and update orders.

Objective 7 — Reviews and Ratings

Allow verified customers to submit product reviews and ratings to help future buyers make informed decisions.

Objective 8 — Inventory Management

Automatically update stock quantities after purchases and notify administrators when inventory reaches low levels.

Objective 9 — Business Analytics

Provide dashboards displaying sales statistics, revenue, product performance, customer growth, and inventory summaries.

Objective 10 — Cloud Integration

Store product images securely using Cloudinary to improve scalability and reduce server storage requirements.

Objective 11 — Email Notifications

Automatically send emails for account verification, password reset, order confirmation, payment confirmation, shipping updates, and delivery notifications.

Objective 12 — Performance

Optimize application performance using techniques such as Redis caching, database indexing, pagination, lazy loading, and image optimization.

Objective 13 — Security

Protect the application using secure authentication, password hashing, input validation, secure HTTP headers, rate limiting, and SQL injection prevention.

Objective 14 — Responsive Design

Ensure the application works seamlessly across desktop, tablet, and mobile devices.

Objective 15 — Deployment

Deploy the application using modern cloud platforms with proper environment configuration, HTTPS, and production-ready settings.

SMART Objectives

Objective	Measure of Success
Authentication	Users can register, verify email, log in, and access authorized pages.
Product Management	Admin can manage products, categories, brands, images, and variants.
Shopping Cart	Customers can add, update, and remove items successfully.
Checkout	Orders can be completed using Stripe without errors.

Objective	Measure of Success
Inventory	Stock updates automatically after successful orders.
Order Management	Customers and admins can view and update order information.
Reviews	Verified customers can submit ratings and reviews.
Performance	Product pages load quickly with optimized queries and caching.
Security	Unauthorized access is blocked, and user data is protected.
Deployment	Frontend, backend, and database are accessible in production.

Expected Outcomes

By the end of the project, the system will provide:

- A complete online shopping platform.
- A responsive customer interface.
- A powerful admin dashboard.
- Secure online payments.
- Automated inventory management.
- Order tracking and notifications.
- Business reports and analytics.
- Production-ready deployment.

Phase 1 — Step 1.4: Project Scope

Purpose

The project scope defines the boundaries of the system. It clearly specifies:

- What features are included in Version 1 (MVP + Production-ready)
- What features are excluded
- Who will use the system
- What technologies and platforms will be supported

---

1. In-Scope Features

A. User Authentication

The system shall allow users to:

- Register an account
- Verify email address
- Log in securely
- Log out
- Reset forgotten passwords
- Update profile information
- Change password
- Manage saved addresses

B. Customer Module

Customers shall be able to:

- Browse all products
- Search products
- Filter by category
- Filter by brand
- Filter by price
- Sort products
- View product details
- View product images
- View related products
- View product ratings
- Add products to cart
- Update cart quantities
- Remove cart items
- Save products to wishlist
- Apply coupons
- Checkout securely
- View order history
- Track order status
- Submit reviews and ratings

C. Admin Module

Administrators shall be able to:

Dashboard

- View total sales
- View revenue
- View orders
- View customers
- View low-stock products
- View recent orders

Product Management

- Add products
- Edit products
- Delete products
- Upload multiple images
- Manage product variants
- Manage inventory

Category Management

- Add categories
- Edit categories
- Delete categories

Brand Management

- Add brands
- Edit brands
- Delete brands

Order Management

- View orders
- Update order status
- Cancel orders
- Refund orders (optional)
- Print invoices

Customer Management

- View customers
- Block/unblock users
- Manage user roles

Coupon Management

- Create coupons
- Edit coupons
- Delete coupons
- Set expiry dates
- Configure discounts

Reports

- Sales reports
- Revenue reports
- Inventory reports
- Customer reports

D. Shopping Cart

The cart system shall support:

- Add items
- Remove items
- Update quantity
- Calculate subtotal
- Calculate discounts
- Calculate shipping
- Calculate tax
- Display grand total

E. Payment Module

The system shall support:

- Stripe payment
- Payment confirmation
- Payment history
- Failed payment handling

F. Email Module

Automatic emails for:

- Registration
- Email verification
- Password reset
- Order confirmation
- Shipping updates
- Delivery confirmation

G. Inventory Management

The system shall:

- Reduce stock after purchase
- Prevent overselling
- Notify low stock
- Track stock quantity

H. Product Reviews

Customers shall:

- Rate products
- Write reviews
- Edit reviews
- Delete reviews

Only verified purchasers can review products.

I. Search Module

Support:

- Keyword search
- Category search
- Brand search
- Price range filtering
- Sorting
- Pagination

J. Dashboard Analytics

Admin dashboard shall display:

- Revenue
- Orders
- Customers
- Best-selling products
- Monthly sales
- Category performance

---

2. Out-of-Scope Features (Version 1)

The following features are intentionally excluded to keep Version 1 manageable:

- Multi-vendor marketplace
- Live chat with sellers
- AI product recommendations
- Loyalty points
- Gift cards
- Cryptocurrency payments
- Auction system
- Voice search
- Multi-language support
- Multi-currency support
- Native Android/iOS applications
- Offline shopping
- Warehouse automation
- Supplier management

- ERP integration
- Advanced recommendation engine
- Machine learning analytics

These can be introduced in Version 2 or later.

---

3. Supported Users

Customer

- Shopping
- Checkout
- Orders
- Reviews

Administrator

- Manage entire platform
- Products
- Orders
- Customers
- Reports

---

4. Supported Platforms

The application will support:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Responsive design for:

- Desktop
- Tablet
- Mobile

---

5. Technology Scope

Frontend

- React
- Vite
- Material UI
- React Router
- Axios
- React Hook Form
- Yup
- TanStack Query

Backend

- Node.js
- Express.js
- MySQL
- JWT
- Bcrypt
- Cloudinary
- Multer
- Nodemailer
- Stripe
- Redis

Deployment

- Vercel
- Render
- Aiven MySQL
- GitHub

---

6. Expected Deliverables

By the end of the project, the following will be completed:

Documentation

- Software Requirements Specification (SRS)
- API Documentation
- Database Schema
- ER Diagram
- UML Diagrams
- Deployment Guide
- README

Backend

- RESTful APIs
- Authentication
- Payment Integration
- Image Upload
- Email Services

Frontend

- Customer Website
- Admin Dashboard
- Responsive UI
- Protected Routes

Deployment

- Live Frontend
- Live Backend
- Live Database
- Cloud Image Storage

Scope Summary

Included	Excluded
Authentication	Multi-vendor marketplace
Product Management	AI recommendations
Cart & Wishlist	Cryptocurrency payments
Stripe Payments	Native mobile apps

Included	Excluded
Orders	ERP integration
Reviews & Ratings	Voice search
Coupons	Multi-language
Inventory	Warehouse automation
Reports & Analytics	Advanced ML analytics

Phase 1 — Step 1.5: Stakeholders

Purpose

Stakeholders are the individuals or groups who **use**, **manage**, **develop**, or **benefit** from the E-Commerce Platform.

Identifying stakeholders ensures that the system meets the needs of everyone involved.

Primary Stakeholders

1. Customer

Description

A customer is the primary user of the platform who purchases products online.

Responsibilities

- Register an account
- Log in
- Browse products
- Search products
- Add products to cart
- Manage wishlist
- Place orders
- Track orders
- Write reviews
- Manage profile and addresses

Goals

- Fast product search
- Secure payments
- Easy checkout
- Reliable order tracking
- Good shopping experience

---

2. Administrator

Description

The administrator has complete control over the platform.

Responsibilities

- Manage products
- Manage categories
- Manage brands
- Manage inventory
- Manage users
- Manage orders
- Create coupons
- View reports
- Configure platform settings

Goals

- Efficient platform management
- Accurate inventory
- Business growth
- Better sales analytics

---

3. Staff (Order Manager)

Description

Staff members assist administrators with day-to-day operations but have limited permissions.

Responsibilities

- View customer orders
- Update order status

- Pack orders
- Mark orders as shipped
- Process returns
- View inventory

Restrictions

Cannot:

- Delete users
- Change system settings
- Manage administrators
- Configure payments
- Manage roles

Goals

- Fast order processing
- Accurate order fulfillment
- Reduced delivery delays

Secondary Stakeholders

4. Store Owner

Description

The business owner monitors overall performance and makes business decisions.

Responsibilities

- Review reports
- Monitor revenue
- Monitor sales trends
- Evaluate business growth

Goals

- Increase profits
- Improve customer satisfaction
- Expand the business

---

5. Payment Gateway

Example: Stripe

Responsibilities

- Process online payments
- Verify transactions
- Handle refunds
- Secure payment data

Goals

- Fast and secure payment processing
- Reliable transaction handling

---

6. Email Service

Examples:

- Gmail (SMTP)
- SendGrid (optional for production)

Responsibilities

- Send verification emails
- Send password reset emails
- Send order confirmation emails
- Send shipping notifications

---

7. Cloud Storage

Example: Cloudinary

Responsibilities

- Store product images
- Optimize image delivery

- Serve images efficiently

---

8. Hosting Providers

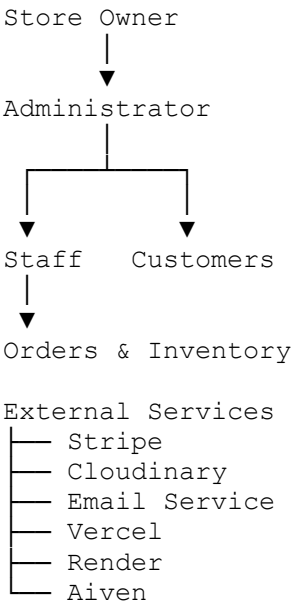
Examples:

- Frontend: Vercel
- Backend: Render
- Database: Aiven

Responsibilities

- Host the application
- Ensure uptime
- Scale resources
- Secure infrastructure

Stakeholder Relationship



Stakeholder Summary

Stakeholder	Role	System Access
Customer	Purchases products	Customer Portal
Administrator	Manages the platform	Admin Dashboard
Staff	Handles orders and inventory	Staff Dashboard
Store Owner	Monitors business	Reports & Analytics
Stripe	Processes payments	Payment APIs
Cloudinary	Stores images	Image APIs
Email Service	Sends notifications	Email APIs
Hosting Providers	Host the application	Infrastructure

Excellent. This is one of the most important steps because it will determine your **JWT authorization**, **database design**, **backend middleware**, and **frontend protected routes**.

Phase 1 — Step 1.6: User Roles & Permissions (RBAC)

Purpose

Role-Based Access Control (RBAC) ensures that users can only access the features and data appropriate for their role.

System Roles

The system will have **three user roles**.

1. Customer
2. Staff (Order Manager)
3. Administrator

---

1. Customer

Description

A customer is a registered user who can browse products, make purchases, manage their account, and track orders.

Customer Permissions

Authentication

- Register
- Login
- Logout
- Verify Email
- Reset Password
- Update Profile
- Change Password

Products

- Browse Products
- Search Products
- Filter Products
- Sort Products
- View Product Details

Shopping

- Add to Cart
- Update Cart
- Remove from Cart
- Clear Cart

Wishlist

- Add to Wishlist
- Remove from Wishlist
- View Wishlist

Orders

- Checkout
- Make Payment
- View Order History
- View Order Details
- Track Orders
- Cancel Order (Before Shipment)

Reviews

- Add Review (Verified Purchase Only)
- Edit Own Review
- Delete Own Review

Address

- Add Address
- Update Address
- Delete Address

Restrictions

Customers **cannot**:

- Access admin dashboard
- Manage products
- Manage categories
- Manage brands
- Manage users
- View other customers' orders
- Access reports
- Manage inventory
- Configure coupons

---

2. Staff (Order Manager)

Description

Staff members assist with daily operations but do not have full administrative privileges.

Staff Permissions

Dashboard

- View Staff Dashboard

Orders

- View Orders
- Update Order Status
- Mark as Packed
- Mark as Shipped
- Mark as Delivered
- Process Returns

Inventory

- View Inventory
- Update Stock Quantity

Customers

- View Customer Information

Reports

- View Daily Orders
- View Shipment Reports

Restrictions

Staff **cannot**:

- Delete Products
- Delete Users
- Create Administrators
- Manage Roles
- Configure Stripe
- Create Coupons
- Access Security Settings
- Delete Orders

---

3. Administrator

Description

Administrators have full control over the application.

Dashboard

- View Analytics
- View Revenue
- View Reports
- View Statistics

Product Management

- Create Products
- Edit Products
- Delete Products
- Upload Images
- Manage Variants
- Manage Categories
- Manage Brands

Inventory

- Manage Inventory
- Manage Stock
- View Low Stock Alerts

Orders

- View Orders
- Update Status
- Cancel Orders
- Refund Orders
- Print Invoices

Customers

- View Customers
- Block Users
- Unblock Users
- Manage Roles

Coupons

- Create Coupons
- Update Coupons
- Delete Coupons

Reports

- Revenue Reports
- Product Reports
- Customer Reports
- Inventory Reports

Settings

- Configure Platform
- Configure Payment Gateway
- Configure Email Service

Administrator Restrictions

Only actions protected by business rules, for example:

- Cannot delete their own account while logged in.
- Cannot permanently delete completed financial records.
- Cannot modify historical payment transaction data.

Permission Matrix

Feature	Customer	Staff	Admin
Register	✓	✗	✗
Login	✓	✓	✓
Browse Products	✓	✓	✓
Search Products	✓	✓	✓
Add to Cart	✓	✗	✗
Wishlist	✓	✗	✗
Checkout	✓	✗	✗
Stripe Payment	✓	✗	✗
View Own Orders	✓	✗	✗
View All Orders	✗	✓	✓
Update Order Status	✗	✓	✓
Manage Products	✗	✗	✓
Manage Categories	✗	✗	✓
Manage Brands	✗	✗	✓
Manage Inventory	✗	✓	✓
Manage Users	✗	✗	✓
Manage Coupons	✗	✗	✓
View Reports	✗	Limited	✓

Feature	Customer	Staff	Admin
Configure Settings	✗	✗	✓

Authorization Middleware

Your backend will use middleware to protect routes.

Examples:

```
Public Routes
GET    /products
POST   /auth/login
POST   /auth/register

Customer Routes
POST   /cart
POST   /checkout
GET    /orders/my

Staff Routes
GET    /staff/orders
PATCH  /staff/orders/:id/status

Admin Routes
POST   /admin/products
PUT    /admin/products/:id
DELETE /admin/products/:id
GET    /admin/reports
```

JWT Flow

1. User logs in.
2. Server verifies credentials.
3. Server generates a JWT containing:
    - User ID
    - Role
4. Client stores the token securely.
5. Every protected request includes the token.
6. Middleware validates the token.
7. Middleware checks the user's role before allowing access.

Phase 1 — Step 1.7: Functional Requirements

Purpose

Functional requirements describe **what the system must do**. These requirements will directly translate into your backend APIs, frontend pages, database tables, and business logic.

FR-1 Authentication Module

FR-1.1 User Registration

The system shall allow new users to create an account by providing:

- First Name
- Last Name
- Email
- Password
- Confirm Password

Validation

- Email must be unique.
- Password must meet the password policy.
- Required fields cannot be empty.

FR-1.2 Login

The system shall authenticate users using:

- Email
- Password

Upon successful login:

- Generate JWT Access Token
- Generate Refresh Token (optional)
- Redirect users based on role

FR-1.3 Logout

The system shall invalidate the current session and remove authentication tokens.

FR-1.4 Forgot Password

The system shall:

- Accept user's email.
- Generate a secure reset token.
- Send a password reset email.
- Allow password reset before token expiration.

FR-1.5 Email Verification

The system shall:

- Send a verification email after registration.
- Prevent unverified users from placing orders (recommended).
- Activate the account after successful verification.

FR-2 User Profile Module

The customer shall be able to:

- View profile
- Edit profile
- Change password
- Upload profile picture (optional)
- Manage shipping addresses

FR-3 Product Module

Product Management

Administrator shall:

- Add products
- Edit products
- Delete products
- Upload multiple images
- Assign categories
- Assign brands
- Manage product variants
- Manage stock
- Set pricing

Product Display

Customer shall:

- View product list
- View product details
- View images
- View variants
- View availability
- View ratings
- View reviews

FR-4 Search Module

Customers shall be able to search using:

- Product name
- Category
- Brand
- SKU (optional)

FR-5 Filtering Module

Customers shall filter by:

- Category
- Brand
- Price Range
- Rating
- Availability

FR-6 Sorting Module

Sort products by:

- Newest
- Price Low to High
- Price High to Low
- Highest Rated
- Best Selling
- Alphabetical (A–Z)

FR-7 Shopping Cart Module

Customers shall:

- Add products
- Remove products
- Update quantity
- Clear cart
- View subtotal
- View tax
- View shipping
- View total

FR-8 Wishlist Module

Customers shall:

- Add products

- Remove products
- Move items to cart
- View wishlist

FR-9 Coupon Module

Customers shall:

- Apply coupon
- Remove coupon

System shall:

- Validate coupon
- Check expiration
- Check usage limits
- Calculate discount

FR-10 Checkout Module

Customers shall:

- Select shipping address
- Select payment method
- Review order summary
- Confirm purchase

System shall:

- Calculate totals
- Generate invoice
- Create order
- Update inventory

FR-11 Payment Module

The system shall integrate with Stripe to:

- Create payment session
- Process payment
- Verify payment
- Record transaction
- Handle payment failure
- Support refunds (admin)

FR-12 Order Module

Customers shall:

- View order history
- View order details
- Track order status
- Cancel eligible orders

Staff/Admin shall:

- View all orders
- Update status
- Process returns
- Generate invoices

FR-13 Review Module

Verified customers shall:

- Add review
- Edit review
- Delete review
- Rate products (1–5 stars)

System shall:

- Calculate average rating
- Display review count

FR-14 Inventory Module

System shall:

- Reduce stock after successful payment
- Prevent purchases beyond available stock
- Notify low inventory

Admin shall:

- Update stock
- View stock history (optional)

FR-15 Category Module

Administrator shall:

- Create category
- Update category
- Delete category
- Activate/deactivate category

FR-16 Brand Module

Administrator shall:

- Create brand
- Update brand
- Delete brand
- Upload brand logo (optional)

FR-17 Dashboard Module

Customer Dashboard

- Profile summary

- Recent orders
- Saved addresses
- Wishlist

Staff Dashboard

- Orders to process
- Orders shipped today
- Low stock alerts

Admin Dashboard

- Revenue
- Orders
- Customers
- Products
- Sales charts
- Top-selling products
- Recent activities

FR-18 Reports Module

Admin shall generate reports for:

- Sales
- Orders
- Products
- Customers
- Inventory
- Coupons

Reports may support filtering by date range.

FR-19 Email Module

The system shall automatically send emails for:

- Welcome message
- Email verification

- Password reset
- Order confirmation
- Payment confirmation
- Shipping update
- Delivery confirmation

FR-20 Security Module

The system shall:

- Hash passwords
- Validate input
- Authorize requests using JWT
- Restrict access by role
- Prevent unauthorized API access
- Log security-related events (optional)

Functional Requirements Summary

Module	Core Functionality
Authentication	Register, login, logout, password reset
User Profile	Manage profile and addresses
Products	CRUD, variants, images, stock
Search & Filter	Product discovery
Cart & Wishlist	Shopping management
Coupons	Discounts and validation
Checkout	Order creation and confirmation
Payments	Stripe integration
Orders	Tracking and management
Reviews	Ratings and feedback
Inventory	Stock management
Categories & Brands	Catalog organization
Dashboards	Customer, staff, admin views
Reports	Sales and business insights
Emails	Notifications
Security	Authentication and authorization

Excellent. Now we'll write the **Non-Functional Requirements (NFRs)**. These define **how well the system should perform**, rather than what it should do. This is an essential part of a professional SRS.

Phase 1 — Step 1.8: Non-Functional Requirements (NFR)

Purpose

Non-functional requirements define the quality attributes of the system, such as performance, security, reliability, usability, and scalability.

NFR-1 Performance

NFR-PER-001 — Page Load Time

The system should load most pages within **3 seconds** under normal network conditions.

NFR-PER-002 — API Response Time

Most API requests should respond within:

- **Read Operations:** ≤ 500 ms
- **Write Operations:** ≤ 1 second

NFR-PER-003 — Search Performance

Product searches should return results within **2 seconds**.

NFR-PER-004 — Pagination

Large datasets shall use server-side pagination to maintain performance.

NFR-2 Scalability

NFR-SCA-001

The application should support future expansion without major architectural changes.

Examples:

- Additional user roles
- Multi-vendor marketplace
- Mobile application
- Multi-language support

NFR-SCA-002

The backend shall expose RESTful APIs to allow future integrations.

NFR-3 Security

NFR-SEC-001

Passwords shall be securely hashed using **bcrypt** before storage.

NFR-SEC-002

Protected endpoints shall require valid JWT authentication.

NFR-SEC-003

Role-Based Access Control (RBAC) shall restrict access to authorized users only.

NFR-SEC-004

The application shall validate and sanitize all user inputs.

NFR-SEC-005

Sensitive configuration values (API keys, database credentials, JWT secrets) shall be stored in environment variables and never hard-coded.

NFR-SEC-006

The application should implement rate limiting on authentication endpoints to reduce brute-force attacks.

NFR-4 Reliability

NFR-REL-001

The application should handle unexpected errors gracefully without crashing.

NFR-REL-002

Failed transactions shall not leave inconsistent data in the database.

NFR-REL-003

Critical application errors should be logged for debugging and monitoring.

NFR-5 Availability

NFR-AVA-001

The deployed application should aim for **99% or higher availability**.

NFR-AVA-002

Users should receive meaningful error messages during temporary service interruptions.

NFR-6 Usability

NFR-USA-001

The user interface shall be intuitive and easy to navigate.

NFR-USA-002

The application shall be responsive on:

- Desktop
- Tablet
- Mobile

NFR-USA-003

Forms shall display clear validation messages when user input is invalid.

NFR-7 Maintainability

NFR-MAI-001

The project shall follow a modular architecture (MVC for backend, component-based architecture for frontend).

NFR-MAI-002

Source code shall follow consistent naming conventions and formatting standards.

NFR-MAI-003

Business logic should be separated from routing and presentation logic.

NFR-8 Compatibility

NFR-COM-001

The application shall support the latest versions of:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

NFR-9 Database

NFR-DB-001

The database shall enforce:

- Primary Keys
- Foreign Keys
- Unique Constraints
- Appropriate Indexes

NFR-DB-002

The database design shall comply with **Third Normal Form (3NF)** where appropriate.

NFR-10 Backup & Recovery

NFR-BKP-001

Production databases should be backed up regularly using the hosting provider's backup capabilities.

NFR-BKP-002

The application should support recovery from backups in case of data loss.

NFR-11 Logging & Monitoring

NFR-LOG-001

The application shall log:

- Authentication failures
- Server errors
- Payment failures
- Unexpected exceptions

NFR-LOG-002

Logs shall not expose sensitive user information such as passwords or payment details.

NFR-12 Deployment

NFR-DEP-001

The frontend, backend, and database shall be deployable independently.

NFR-DEP-002

Configuration shall be managed through environment variables to support multiple environments (development, testing, production).

NFR-13 Accessibility

NFR-ACC-001

The application should use semantic HTML where appropriate.

NFR-ACC-002

Interactive elements should be accessible using a keyboard.

NFR-ACC-003

Images should include descriptive alternative text where applicable.

NFR-14 Documentation

NFR-DOC-001

The project shall include:

- README
- API Documentation
- Database Schema
- SRS
- Deployment Guide

Non-Functional Requirements Summary

Category	Goal
Performance	Fast page loads and API responses
Scalability	Support future growth and integrations
Security	Protect user data and system access
Reliability	Stable operation and data consistency
Availability	High system uptime
Usability	Responsive and user-friendly interface
Maintainability	Modular, clean, and organized code
Compatibility	Support modern web browsers
Database	Normalized and optimized schema
Backup & Recovery	Protect against data loss
Logging	Aid monitoring and debugging
Deployment	Flexible cloud deployment
Accessibility	Improve usability for all users
Documentation	Complete project documentation

Phase 1 — Step 1.9: Business Rules

Purpose

Business Rules define the policies and constraints that the E-Commerce Platform must enforce to ensure correct and consistent business operations.

BR-1 User Management

BR-001

Each user must register using a **unique email address**.

BR-002

Passwords must satisfy the system's password policy.

Example:

- Minimum 8 characters
- At least one uppercase letter
- At least one lowercase letter
- At least one number
- At least one special character

BR-003

Only verified users can place orders.

BR-004

Users can only edit their own profile.

BR-005

Users cannot access another user's private information.

BR-2 Product Management

BR-006

Each product must belong to **one category**.

BR-007

Each product must belong to **one brand**.

BR-008

Every product must have:

- Name
- Price
- Stock
- Category
- Brand

BR-009

Product price cannot be negative.

BR-010

Stock quantity cannot be negative.

BR-011

Inactive products shall not be displayed to customers.

BR-3 Product Variants

BR-012

Variants (Size, Color, etc.) belong to a specific product.

BR-013

Each variant maintains its own stock quantity.

BR-014

Customers must select required variants before adding products to the cart.

BR-4 Shopping Cart

BR-015

A customer cannot add more items than the available stock.

BR-016

Cart quantities must always remain greater than zero.

BR-017

Removing the final item results in an empty cart.

BR-018

The cart total is recalculated whenever products, quantities, coupons, or shipping costs change.

BR-5 Wishlist

BR-019

A product cannot appear more than once in a customer's wishlist.

BR-6 Coupon Rules

BR-020

Coupons must have an expiration date.

BR-021

Expired coupons cannot be applied.

BR-022

Disabled coupons cannot be used.

BR-023

Coupons may have:

- Minimum purchase amount
- Maximum discount amount
- Usage limits

BR-024

Only one coupon can be applied to an order.

BR-7 Checkout

BR-025

Checkout requires at least one product in the cart.

BR-026

A valid shipping address is required before checkout.

BR-027

The final order amount equals:

Subtotal − Discount + Shipping + Tax

BR-8 Payments

BR-028

Orders are created only after successful payment confirmation (for online payments).

BR-029

Failed payments shall not reduce inventory.

BR-030

Each successful payment generates one payment record.

BR-031

Payment records cannot be modified after confirmation.

BR-9 Inventory

BR-032

Inventory is reduced only after successful order confirmation.

BR-033

Products with zero stock cannot be purchased.

BR-034

Low-stock alerts are generated when stock falls below the configured threshold.

BR-10 Orders

BR-035

Each order must have a unique order number.

BR-036

Order status flow:

```
Pending
   ↓
Paid
   ↓
Processing
   ↓
Packed
   ↓
Shipped
   ↓
Delivered
```

Alternative outcomes:

```
Pending → Cancelled
```

```
Paid → Refunded
```

BR-037

Delivered orders cannot be cancelled.

BR-038

Cancelled orders restore inventory if stock was already reserved or deducted.

BR-11 Reviews

BR-039

Only customers who purchased a product may submit a review.

BR-040

Each customer may submit only one review per purchased product.

BR-041

Ratings must be between **1 and 5 stars**.

BR-12 Administration

BR-042

Only Administrators can:

- Manage products
- Manage categories
- Manage brands
- Manage coupons
- Manage users
- Configure system settings

BR-043

Staff members may update order status but cannot manage users, products, or platform settings.

BR-13 Reports

BR-044

Revenue reports include only successfully paid orders.

BR-045

Cancelled orders are excluded from revenue calculations.

BR-14 Security

BR-046

Users may only access resources permitted by their role.

BR-047

Sensitive data such as passwords must never be stored in plain text.

BR-048

JWT tokens must be validated before accessing protected APIs.

BR-15 Audit Trail

BR-049

Important administrative actions should be logged, such as:

- Product creation
- Product deletion
- Price changes
- Coupon creation
- Role changes

Business Rules Summary

Category	Rule Examples
Users	Unique email, verified accounts, own profile only
Products	Category, brand, positive price and stock
Variants	Required selection, separate inventory
Cart	Stock validation, automatic total calculation
Coupons	Expiration, usage limits, one coupon per order
Checkout	Valid cart, shipping address required
Payments	Payment confirmation before order completion
Inventory	Prevent overselling, low-stock alerts
Orders	Status workflow, cancellation rules
Reviews	Verified purchase only, one review per product
Administration	Role-based management
Reports	Paid orders only for revenue
Security	JWT, RBAC, password hashing
Audit	Log critical administrative actions

Phase 1 — Step 1.10: Assumptions & Constraints

Purpose

This section documents:

- Assumptions made during development.
- Technical limitations.
- Business limitations.

- Project constraints.

These help everyone understand the project's boundaries.

Part A — Assumptions

ASS-001 Internet Connectivity

The application assumes that users have a stable internet connection while browsing, placing orders, and making payments.

ASS-002 Browser Support

Users are assumed to use a modern web browser that supports current web standards.

Examples:

- Chrome
- Edge
- Firefox
- Safari

ASS-003 Email Availability

Users are assumed to have access to a valid email address for:

- Registration
- Email verification
- Password recovery
- Order notifications

ASS-004 Payment Gateway Availability

The Stripe payment service is assumed to be available whenever customers perform online payments.

ASS-005 Cloud Storage Availability

Cloudinary is assumed to be available for uploading and serving product images.

ASS-006 Database Availability

The MySQL database is assumed to be online and reachable by the backend server.

ASS-007 Accurate Product Data

Administrators are responsible for maintaining accurate product information, including prices, stock, descriptions, and images.

ASS-008 Authorized Users

All authenticated users are assumed to use their own accounts and not share credentials.

ASS-009 Secure Environment

Production deployment assumes that:

- HTTPS is enabled.
- Environment variables are configured correctly.
- Secrets are protected.

ASS-010 Third-Party Services

The system assumes that third-party services (Stripe, Cloudinary, Email Provider) operate according to their published APIs.

Part B — Constraints

CON-001 Technology Stack

The project will use:

Frontend

- React
- Material UI
- Vite

Backend

- Node.js
- Express.js

Database

- MySQL

Authentication

- JWT

Image Storage

- Cloudinary

Payment

- Stripe

CON-002 Version 1 Scope

Version 1 does **not** include:

- Multi-vendor marketplace
- Mobile applications
- Multi-language support
- Multi-currency support
- AI recommendations

CON-003 Hosting

The application will be deployed using:

- Frontend → Vercel
- Backend → Render
- Database → Aiven MySQL

CON-004 Budget

The project should primarily rely on free or developer-tier services during development and testing.

CON-005 Browser Dependency

The application requires JavaScript to be enabled in the user's browser.

CON-006 Payment Method

Version 1 supports Stripe for online card payments. Additional payment methods can be added in future releases.

CON-007 Image Storage

Product images will be stored in Cloudinary rather than on the application server.

CON-008 Database

The application will use a relational database (MySQL). Switching to another database (e.g., PostgreSQL) would require configuration changes and testing.

CON-009 Development Time

The project is intended to be completed within approximately **10–14 weeks**, depending on feature depth and testing.

CON-010 Team Size

This version is designed and implemented by a single developer, so planning and scope reflect solo development.

Risks (Optional but Recommended)

Although not always included in an SRS, identifying risks is a good engineering practice.

Risk	Possible Impact	Mitigation
Stripe service outage	Payments unavailable	Display clear error messages and allow users to retry later
Cloudinary unavailable	Image uploads fail	Validate uploads and retry or notify the admin
Database downtime	System unavailable	Use managed backups and monitoring
Large image uploads	Slow performance	Compress images and enforce size limits
High traffic	Slower response times	Add caching, optimize queries, and scale infrastructure

---

Assumptions & Constraints Summary

Category	Examples
Assumptions	Internet, email, browser, Stripe, Cloudinary, MySQL availability
Technical Constraints	React, Express, MySQL, JWT
Business Constraints	Version 1 feature scope
Infrastructure Constraints	Vercel, Render, Aiven
Development Constraints	Solo developer, limited budget, project timeline

---

Phase 1 — Step 1.11: User Stories

Purpose

User stories describe the features from the perspective of the end users. They help developers understand user needs and later become tasks in a development backlog.

Customer User Stories

Authentication

US-CUST-001

As a customer, I want to create an account so that I can purchase products and track my orders.

US-CUST-002

As a customer, I want to log in securely so that I can access my account.

US-CUST-003

As a customer, I want to reset my password so that I can regain access if I forget it.

US-CUST-004

As a customer, I want to verify my email so that my account is secure.

Profile

US-CUST-005

As a customer, I want to update my profile information so that my account details remain accurate.

US-CUST-006

As a customer, I want to manage multiple shipping addresses so that I can ship orders to different locations.

Product Browsing

US-CUST-007

As a customer, I want to browse all products so that I can discover items to purchase.

US-CUST-008

As a customer, I want to search products by keyword so that I can quickly find specific items.

US-CUST-009

As a customer, I want to filter products by category, brand, price, and rating so that I can narrow my search.

US-CUST-010

As a customer, I want to sort products by price, popularity, and newest arrivals so that I can find products more efficiently.

US-CUST-011

As a customer, I want to view product details, images, specifications, and reviews so that I can make informed purchasing decisions.

Shopping Cart

US-CUST-012

As a customer, I want to add products to my shopping cart so that I can purchase them later.

US-CUST-013

As a customer, I want to update product quantities in my cart so that I can adjust my order before checkout.

US-CUST-014

As a customer, I want to remove products from my cart so that I can manage my intended purchases.

US-CUST-015

As a customer, I want to see the subtotal, discounts, shipping, tax, and total cost so that I know the final amount before payment.

Wishlist

US-CUST-016

As a customer, I want to save products to my wishlist so that I can purchase them later.

US-CUST-017

As a customer, I want to move items from my wishlist to my cart so that I can complete my purchase.

Checkout & Payment

US-CUST-018

As a customer, I want to apply discount coupons so that I can reduce my order cost.

US-CUST-019

As a customer, I want to pay securely using Stripe so that my payment information remains protected.

US-CUST-020

As a customer, I want to receive an order confirmation after payment so that I know my order was successful.

Orders

US-CUST-021

As a customer, I want to track my order status so that I know when it will arrive.

US-CUST-022

As a customer, I want to view my order history so that I can review previous purchases.

US-CUST-023

As a customer, I want to cancel eligible orders so that I can change my mind before shipment.

Reviews

US-CUST-024

As a customer, I want to submit reviews and ratings for purchased products so that I can share my experience with others.

US-CUST-025

As a customer, I want to edit my review so that I can correct or update my feedback.

Staff User Stories

US-STAFF-001

As a staff member, I want to view pending orders so that I can begin processing them.

US-STAFF-002

As a staff member, I want to update order statuses so that customers receive accurate progress information.

US-STAFF-003

As a staff member, I want to process returns so that returned products are handled correctly.

US-STAFF-004

As a staff member, I want to update inventory quantities so that stock information remains accurate.

US-STAFF-005

As a staff member, I want to view low-stock products so that I can notify administrators.

Administrator User Stories

US-ADMIN-001

As an administrator, I want to create products so that new items are available for purchase.

US-ADMIN-002

As an administrator, I want to edit product information so that listings remain accurate and up to date.

US-ADMIN-003

As an administrator, I want to delete products so that discontinued items are removed from the catalog.

US-ADMIN-004

As an administrator, I want to manage categories and brands so that products are organized properly.

US-ADMIN-005

As an administrator, I want to upload multiple product images so that customers can view products from different angles.

US-ADMIN-006

As an administrator, I want to manage coupons so that promotional campaigns can be offered to customers.

US-ADMIN-007

As an administrator, I want to manage users and roles so that platform access remains secure and organized.

US-ADMIN-008

As an administrator, I want to manage inventory so that stock levels remain accurate.

US-ADMIN-009

As an administrator, I want to view dashboards and analytics so that I can monitor business performance.

US-ADMIN-010

As an administrator, I want to generate sales and inventory reports so that I can make informed business decisions.

US-ADMIN-011

As an administrator, I want to view all payments so that financial transactions can be monitored.

US-ADMIN-012

As an administrator, I want to configure system settings such as payment and email services so that the platform functions correctly.

User Story Summary

Role	Stories
Customer	25
Staff	5
Administrator	12
Total	**42**

Product Backlog (High-Level)

Priority	Module
High	Authentication
High	Product Catalog
High	Shopping Cart
High	Checkout
High	Payment
High	Orders
Medium	Wishlist
Medium	Reviews
Medium	Coupons
Medium	Reports
Low	Advanced Analytics
Low	Future Features

Sprint Plan (Suggested)

Sprint 1

- Authentication
- User Profile

- RBAC

Sprint 2

- Products
- Categories
- Brands
- Inventory

Sprint 3

- Search
- Filters
- Pagination
- Wishlist

Sprint 4

- Cart
- Coupons
- Checkout

Sprint 5

- Stripe Integration
- Orders
- Email Notifications

Sprint 6

- Reviews
- Reports
- Dashboard
- Deployment

Phase 1 — Step 1.12: Use Case Diagrams

Purpose

A Use Case Diagram visually shows:

- Actors (who interacts with the system)

- Use Cases (what they can do)
- Relationships between actors and use cases

These diagrams help developers, testers, and stakeholders understand the system's functionality.

UML Package Structure

We will create **five** use case diagrams.

```
Use Case Diagrams
│
├── UC-01 System Overview
├── UC-02 Authentication
├── UC-03 Customer Module
├── UC-04 Staff Module
└── UC-05 Admin Module
```

UC-01 — System Overview

Actors

- Customer
- Staff
- Administrator
- Stripe (External System)
- Cloudinary (External System)
- Email Service (External System)

High-Level Use Cases

Customer

- Register
- Login
- Browse Products
- Search Products
- Manage Cart
- Manage Wishlist
- Checkout
- Make Payment

- View Orders
- Track Orders
- Submit Reviews

Staff

- Login
- View Orders
- Update Order Status
- Manage Inventory

Administrator

- Login
- Manage Products
- Manage Categories
- Manage Brands
- Manage Coupons
- Manage Customers
- Manage Orders
- View Reports
- View Dashboard
- Configure Settings

External Systems

Stripe

- Process Payment
- Verify Payment
- Refund Payment

Cloudinary

- Upload Product Images
- Delete Product Images

Email Service

- Send Verification Email
- Send Password Reset Email
- Send Order Confirmation
- Send Shipping Updates

UC-02 — Authentication Module

Actors

- Customer
- Staff
- Administrator

Use Cases

- Register
- Login
- Logout
- Verify Email
- Forgot Password
- Reset Password
- Change Password
- Update Profile

Relationships

```
Login
   │
   ├── Validate Credentials
   ├── Generate JWT
   └── Redirect by Role
```

UC-03 — Customer Module

Customer Use Cases

Product

- Browse Products
- Search Products
- Filter Products
- Sort Products
- View Product Details

Shopping

- Add to Cart
- Update Cart
- Remove from Cart
- View Cart

Wishlist

- Add to Wishlist
- Remove from Wishlist
- Move to Cart

Checkout

- Apply Coupon
- Select Address
- Choose Payment
- Place Order

Orders

- View Order History
- Track Orders
- Cancel Order

Reviews

- Add Review
- Edit Review
- Delete Review

UC-04 — Staff Module

Staff Use Cases

Orders

- View Orders
- Update Order Status
- Process Returns

Inventory

- View Inventory
- Update Stock

Dashboard

- View Assigned Orders
- View Low Stock Alerts

UC-05 — Admin Module

Product Management

- Add Product
- Edit Product
- Delete Product

Category Management

- Add Category
- Edit Category
- Delete Category

Brand Management

- Add Brand
- Edit Brand
- Delete Brand

Coupon Management

- Create Coupon
- Update Coupon
- Delete Coupon

User Management

- View Users
- Block Users
- Assign Roles

Order Management

- View Orders
- Update Orders
- Refund Orders

Reports

- Sales Reports

- Inventory Reports
- Revenue Reports
- Customer Reports

Settings

- Payment Settings
- Email Settings
- Platform Configuration

Include & Extend Relationships

<<include>>

These use cases always occur as part of another use case.

Main Use Case	Included Use Case
Register	Validate Input
Register	Hash Password
Register	Send Verification Email
Login	Validate Credentials
Login	Generate JWT
Checkout	Calculate Total
Checkout	Validate Stock
Checkout	Create Order
Payment	Verify Payment
Payment	Save Transaction

<<extend>>

These use cases occur only under certain conditions.

Base Use Case	Extended Use Case
Checkout	Apply Coupon
Checkout	Use Saved Address

Base Use Case	Extended Use Case
Order	Cancel Order
Product	Submit Review
Payment	Refund Payment
Inventory	Low Stock Alert

Actor Permissions Overview

Use Case	Customer	Staff	Admin
Register	✓	✗	✗
Login	✓	✓	✓
Browse Products	✓	✓	✓
Add to Cart	✓	✗	✗
Checkout	✓	✗	✗
View Orders	Own	All	All
Update Order Status	✗	✓	✓
Manage Products	✗	✗	✓
Manage Categories	✗	✗	✓
Manage Coupons	✗	✗	✓
View Reports	✗	Limited	Full
Configure Settings	✗	✗	✓

Recommended Tools

I recommend using **PlantUML** because:

- It integrates well with VS Code.
- Diagrams are version-controlled as text.
- It's widely used in professional environments.
- You can regenerate diagrams whenever requirements change.

Example project structure:

```
docs/
├── SRS.pdf
├── uml/
│   ├── use-case-system.puml
```

```
│   ├── use-case-authentication.puml
│   ├── use-case-customer.puml
│   ├── use-case-staff.puml
│   └── use-case-admin.puml
└── diagrams/
```

Excellent. This is one of the most valuable UML diagrams because it describes **how a process flows from start to finish**.

Unlike the Use Case Diagram (which shows *what* users can do), the **Activity Diagram** shows **how** each process is executed, including decisions, validations, and alternate paths.

Phase 1 — Step 1.13: Activity Diagrams

Purpose

Activity diagrams model the workflow of important business processes.

They help developers understand:

- Process flow
- Decision points
- Validations
- Success paths
- Failure paths

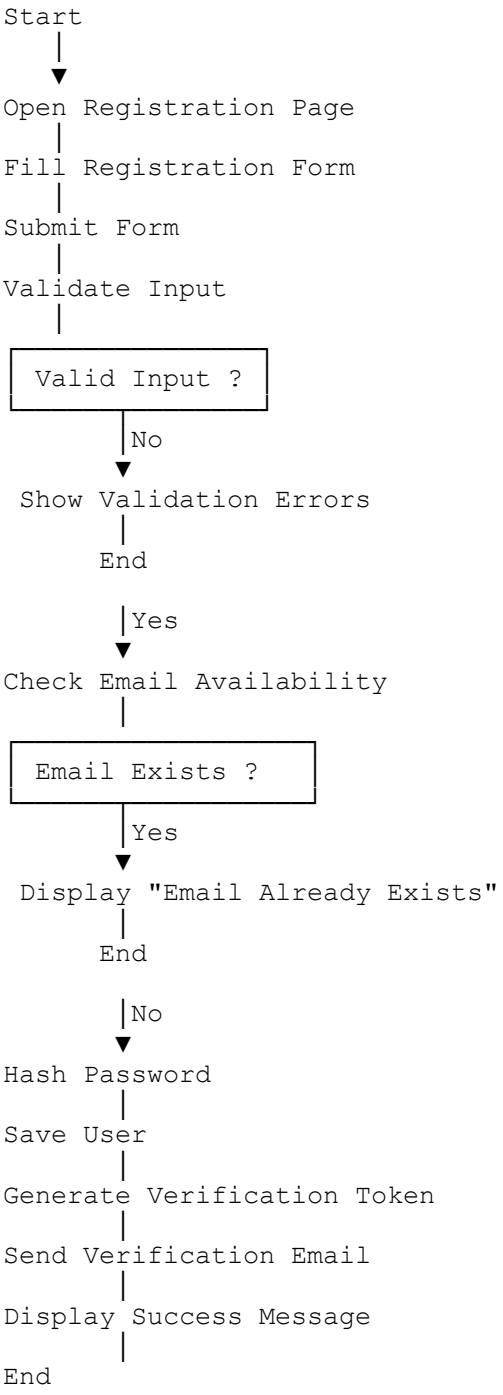
Activity Diagram Package

We will create **six** activity diagrams.

```
Activity Diagrams
│
├── AD-01 User Registration
├── AD-02 User Login
├── AD-03 Customer Checkout
├── AD-04 Stripe Payment
├── AD-05 Order Processing
└── AD-06 Product Management
```

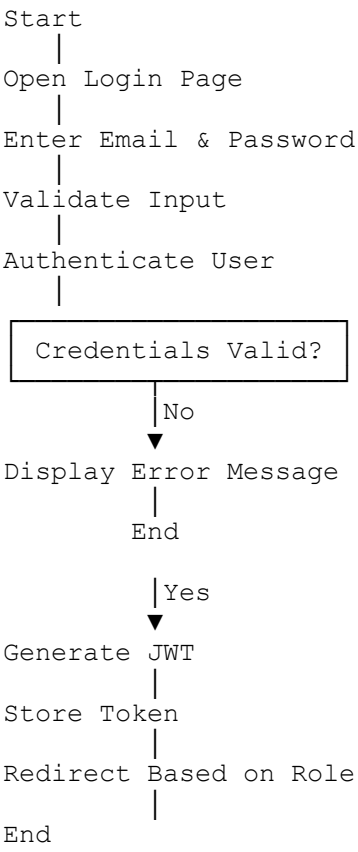
AD-01 — User Registration

Workflow



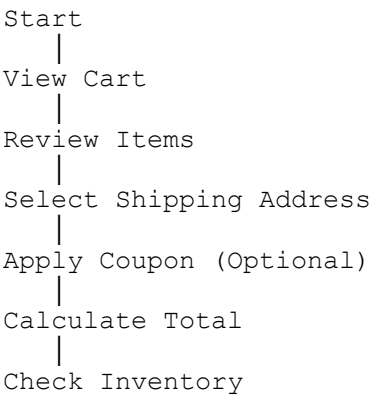
AD-02 — User Login

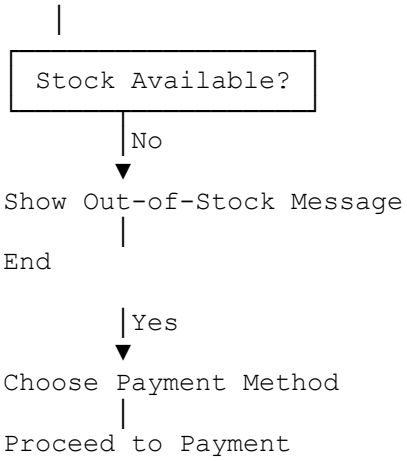
Workflow



AD-03 — Customer Checkout

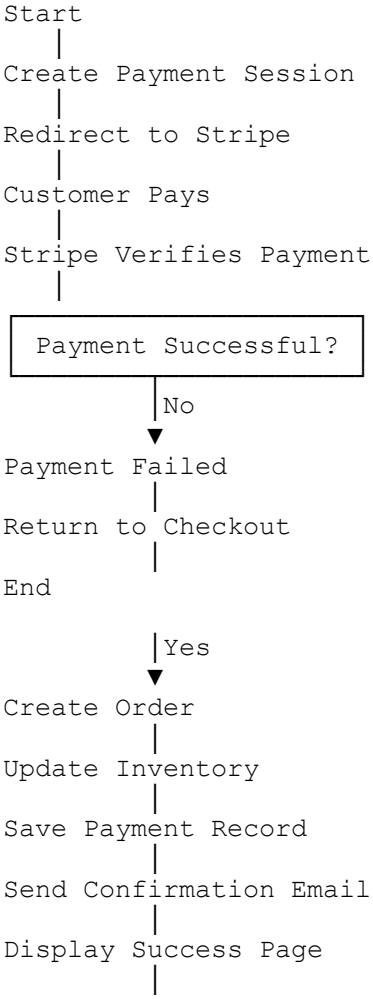
Workflow





AD-04 — Stripe Payment

Workflow



End

AD-05 — Order Processing (Staff/Admin)

Workflow

Start
  |
View New Orders
  |
Accept Order
  |
Pack Items
  |
Update Status → Packed
  |
Ship Order
  |
Update Status → Shipped
  |
Customer Receives Order
  |
Update Status → Delivered
  |
End

AD-06 — Product Management (Admin)

Workflow

Start
  |
Open Product Dashboard
  |
Choose Action
  |

```
Add / Edit / Delete ?
```

Add Product
  |
Upload Images
  |
Enter Product Details
  |
Validate Input
  |

```
Save Product
   |
Display Success

OR

Edit Product
   |
Update Details
   |
Save Changes

OR

Delete Product
   |
Confirm Deletion
   |
Remove Product
   |
Display Success

End
```

Decision Points

Process	Decision
Registration	Email already exists?
Login	Credentials valid?
Checkout	Stock available?
Payment	Payment successful?
Product Management	Add, Edit, or Delete?
Order Processing	Next order status?

Swimlane Recommendation

For clearer diagrams, divide activities into swimlanes.

Registration

Customer	System
Fill form	Validate input
Submit	Check email
	Hash password

Customer	System
	Save user
	Send email

Checkout

Customer	System	Stripe
Review cart	Validate stock	
Confirm order	Create payment session	
Pay		Process payment
	Update order	
	Send confirmation	

Swimlanes make responsibilities immediately clear and are common in professional UML documentation.

Phase 1 — Step 1.14: Sequence Diagrams

Purpose

Sequence diagrams illustrate the order of interactions between system components during a specific process.

Each diagram answers:

- Who starts the process?
- Which component receives the request?
- Which service performs the business logic?
- Which database operations occur?
- What response is returned?

Components Used

Throughout the diagrams, we'll use these participants:

- **User** (Customer, Staff, or Admin)
- **Frontend (React)**
- **Backend API (Express)**
- **Controller**
- **Service**
- **Database (MySQL)**
- **External Service** (Stripe, Cloudinary, Email)

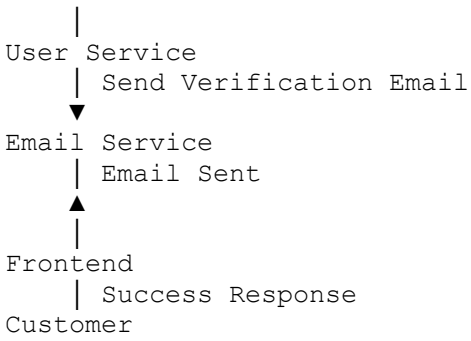
SD-01 — User Registration

Participants

```
Customer
Frontend
Auth API
Auth Controller
User Service
MySQL
Email Service
```

Flow

```
Customer
    │
    ▼
Frontend
    │ Submit Registration
    ▼
Auth API
    │
    ▼
Auth Controller
    │ Validate Input
    ▼
User Service
    │ Check Existing Email
    ▼
MySQL
    │ Email Exists?
    ▲
    │
User Service
    │ Hash Password
    │ Save User
    ▼
MySQL
    │ User Created
    ▲
```

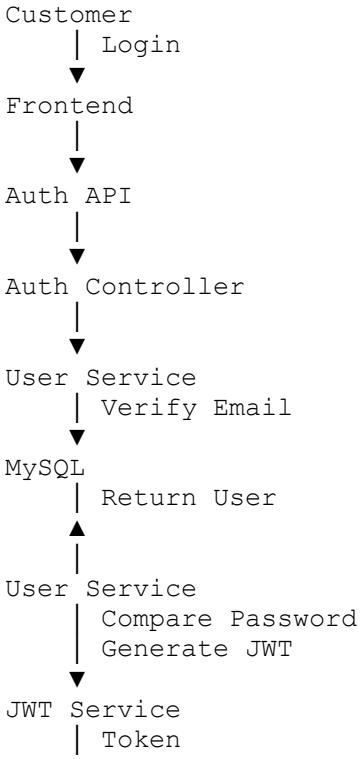


SD-02 — User Login

Participants

Customer
Frontend
Auth API
Auth Controller
User Service
MySQL
JWT Service

Flow



```
    ▲
    |
Frontend
    | Login Success
Customer
```

SD-03 — Product Search

Participants

```
Customer
Frontend
Product API
Product Controller
Product Service
MySQL
```

Flow

```
Customer
    | Search Products
    ▼
Frontend
    | Search Request
    ▼
Product API
    |
    ▼
Controller
    |
    ▼
Service
    | Query Database
    ▼
MySQL
    | Products
    ▲
    |
Frontend
    | Display Results
Customer
```

SD-04 — Checkout

Participants

```
Customer
```

```
Frontend
Order API
Order Controller
Order Service
Cart Service
Inventory Service
MySQL
```

Flow

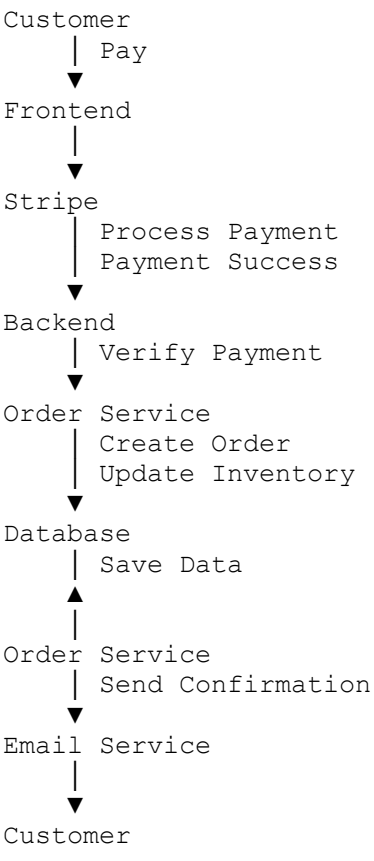
```
Customer
    | Checkout
    ▼
Frontend
    |
    ▼
Order API
    |
    ▼
Order Controller
    |
    ▼
Order Service
    | Validate Cart
    | Validate Stock
    ▼
Inventory Service
    | Stock Available?
    ▲
    |
Order Service
    | Calculate Total
    | Create Payment Session
    ▼
Frontend
```

SD-05 — Stripe Payment

Participants

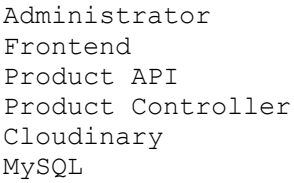
```
Customer
Frontend
Stripe
Backend
Order Service
Inventory
Database
Email Service
```

Flow

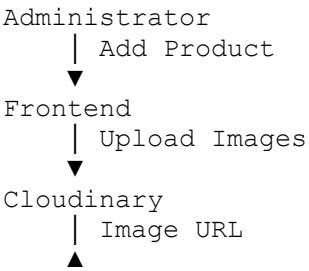


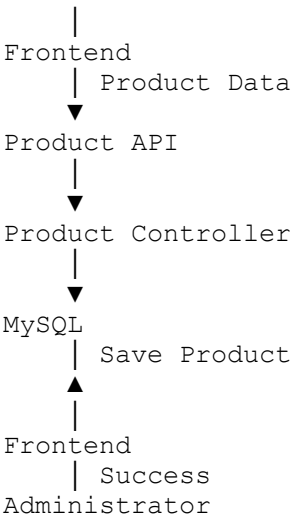
SD-06 — Admin Product Management

Participants



Flow





Sequence Diagram Summary

Diagram	Main Components
Registration	Customer → API → DB → Email
Login	Customer → API → JWT → DB
Product Search	Customer → API → DB
Checkout	Customer → Order Service → Inventory
Payment	Stripe → Backend → DB → Email
Product Management	Admin → Cloudinary → Backend → DB

Design Benefits

These diagrams will directly influence your implementation:

UML Component	Implementation
Frontend	React Pages & API Calls
API	Express Routes
Controller	Controllers
Service	Business Logic Layer
Database	Models & SQL Queries
External Services	Stripe, Cloudinary, Nodemailer

Recommended Project Layers

```
React UI
    |
Axios
    |
Express Routes
    |
Controllers
    |
Services
    |
Models
    |
MySQL
```

This layered architecture keeps business logic out of routes and controllers, making the application easier to test and maintain.

Phase 1 — Step 1.15: High-Level ER Diagram

Purpose

The Entity Relationship Diagram (ERD) defines:

- Database entities (tables)
- Relationships
- Primary Keys (PK)
- Foreign Keys (FK)
- Cardinality
- Database constraints

This ERD will become the foundation for **Phase 2 – Database Design**.

Database Architecture

We'll organize the database into logical modules.

```
Authentication
```

```
│
├── Users
├── Roles
└── UserAddresses

Catalog
│
├── Categories
├── Brands
├── Products
├── ProductImages
└── ProductVariants

Shopping
│
├── Cart
├── CartItems
├── Wishlist
└── WishlistItems

Orders
│
├── Orders
├── OrderItems
├── Payments
└── Coupons

Reviews
│
└── Reviews
```

Entities

1. Roles

```
Role
----
role_id (PK)
name
description
created_at
updated_at
```

---

2. Users

```
User
----
user_id (PK)
role_id (FK)
first_name
```

```
last_name
email
password
phone
profile_image
email_verified
status
created_at
updated_at
```

Relationship

```
Role (1)
      |
      ▼
Users (Many)
```

---

3. User Addresses

```
Address
-------
address_id (PK)
user_id (FK)
country
city
postal_code
street
is_default
created_at
```

Relationship

```
User (1)
    |
    ▼
Addresses (Many)
```

Catalog Module

Categories

```
Category
--------
category_id (PK)
name
```

```
slug
description
status
```

Brands

```
Brand
-----
brand_id (PK)
name
logo
status
```

Products

```
Product
-------
product_id (PK)
category_id (FK)
brand_id (FK)
name
slug
description
price
stock
status
created_at
updated_at
```

Relationship

```
Category (1)
      |
      ▼
Products (Many)

Brand (1)
      |
      ▼
Products (Many)
```

Product Images

```
ProductImage
------------
image_id (PK)
product_id (FK)
image_url
```

```
public_id
```

Relationship

```
Product (1)
      |
      ▼
Images (Many)
```

Product Variants

```
Variant
-------
variant_id (PK)
product_id (FK)
size
color
price_adjustment
stock
```

Relationship

```
Product (1)
      |
      ▼
Variants (Many)
```

Shopping Module

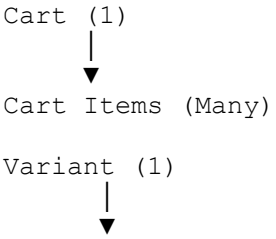
Cart

```
Cart
----
cart_id (PK)
user_id (FK)
created_at
```

Cart Items

```
CartItem
--------
cart_item_id (PK)
cart_id (FK)
variant_id (FK)
quantity
```

Relationship



Wishlist

```
Wishlist
--------
wishlist_id (PK)
user_id (FK)
```

Wishlist Items

```
WishlistItem
------------
wishlist_item_id (PK)
wishlist_id (FK)
product_id (FK)
```

Orders Module

Orders

```
Order
-----
order_id (PK)
user_id (FK)
address_id (FK)
coupon_id (FK)
subtotal
discount
shipping
tax
total
status
created_at
```

Relationship

```
User (1)
     │
     ▼
Orders (Many)
```

Order Items

```
OrderItem
---------
order_item_id (PK)
order_id (FK)
variant_id (FK)
price
quantity
```

Relationship

```
Order (1)
      │
      ▼
Order Items (Many)
```

Payments

```
Payment
-------
payment_id (PK)
order_id (FK)
stripe_payment_id
amount
status
payment_method
paid_at
```

Relationship

```
Order (1)
      │
      ▼
Payment (1)
```

Coupons

```
Coupon
------
```

```
coupon_id (PK)
code
discount_type
discount_value
minimum_purchase
expires_at
status
```

Reviews

```
Review
------
review_id (PK)
user_id (FK)
product_id (FK)
rating
comment
created_at
```

Relationship

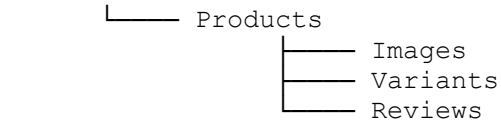
```
User (1)
      |
      ▼
Reviews (Many)

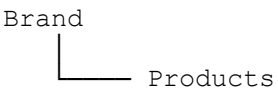
Product (1)
      |
      ▼
Reviews (Many)
```

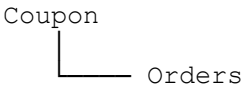
High-Level Relationship Summary

```
Role
 |
 └──── Users
          |
          ├──── Addresses
          ├──── Cart
          |        └──── CartItems
          ├──── Wishlist
          |        └──── WishlistItems
          ├──── Orders
          |        ├──── OrderItems
          |        └──── Payment
          |
          └──── Reviews

Category
      |
```







Cardinality Summary

Parent	Child	Relationship
Role	Users	1 : N
User	Addresses	1 : N
User	Orders	1 : N
User	Reviews	1 : N
User	Cart	1 : 1
User	Wishlist	1 : 1
Category	Products	1 : N
Brand	Products	1 : N
Product	Images	1 : N
Product	Variants	1 : N
Product	Reviews	1 : N
Cart	Cart Items	1 : N
Wishlist	Wishlist Items	1 : N
Order	Order Items	1 : N
Order	Payment	1 : 1

Indexing Strategy

Create indexes for frequently queried columns.

Examples:

- `users.email` (unique)
- `products.slug` (unique)
- `products.category_id`

- `products.brand_id`
- `orders.user_id`
- `orders.status`
- `payments.stripe_payment_id` (unique)
- `coupons.code` (unique)
- `reviews.product_id`

Normalization

The schema is designed to satisfy **Third Normal Form (3NF)**:

- Each table stores a single concept.
- Non-key attributes depend only on the primary key.
- Redundant data is minimized through foreign keys.

Excellent. You've essentially completed the **SRS (Phase 1)**. Before touching the database or writing code, we'll do one final review to make sure the requirements are complete and internally consistent.

Phase 1 — Step 1.16: Final SRS Review & Sign-off

Purpose

The goal of this step is to verify that all requirements, diagrams, and business rules align before implementation begins.

Think of it as the checkpoint where a development team says:

"The requirements are approved. Development can begin."

SRS Checklist

1. Introduction

- ✓ Project Overview
- ✓ Problem Statement
- ✓ Objectives
- ✓ Scope

---

2. Stakeholders

- ✓ Customer
- ✓ Staff
- ✓ Administrator
- ✓ External Systems (Stripe, Cloudinary, Email)

---

3. User Roles

- ✓ Customer
- ✓ Staff
- ✓ Administrator

---

4. Functional Requirements

Examples:

- Authentication
- Product Catalog
- Search
- Cart
- Wishlist
- Checkout
- Payment
- Orders
- Reviews
- Reports
- Dashboard

---

5. Non-Functional Requirements

- Performance
- Security
- Scalability
- Availability
- Maintainability
- Reliability
- Accessibility

---

6. Business Rules

Examples:

- One review per purchased product
- Cannot buy out-of-stock products
- One coupon per order
- Delivered orders cannot be cancelled

---

7. Assumptions & Constraints

- Technology stack
- Hosting
- Budget
- Third-party services
- Version 1 limitations

---

8. User Stories

- Customer
- Staff
- Administrator

---

9. UML

Completed:

- ✓ Use Case Diagrams

- ✓ Activity Diagrams
- ✓ Sequence Diagrams
- ✓ High-Level ER Diagram

Traceability Matrix

Every feature should map through the design.

Requirement	User Story	Use Case	Activity	Sequence	Database
Login	US-CUST-002	Authentication	Login	Login	Users
Search Products	US-CUST-008	Customer	Product Search	Product Search	Products
Checkout	US-CUST-018	Customer	Checkout	Checkout	Orders
Payment	US-CUST-019	Payment	Stripe Payment	Stripe Payment	Payments
Review	US-CUST-024	Customer	Review	Review Flow	Reviews

This matrix ensures nothing is missing between requirements and implementation.

Version 1 Feature Freeze

To keep the project achievable, we'll include only these features in Version 1.

Authentication

- Register
- Login
- Email Verification
- Forgot Password
- JWT Authentication
- RBAC

Catalog

- Categories
- Brands
- Products

- Product Variants
- Product Images

Shopping

- Cart
- Wishlist
- Search
- Filters
- Pagination

Checkout

- Address Management
- Coupons
- Stripe Payment

Orders

- Order History
- Order Tracking
- Order Status History

Inventory

- Stock Management
- Inventory Transactions

Reviews

- Product Reviews
- Product Ratings

Dashboard

Admin

- Dashboard
- Reports
- Product Management
- User Management

Staff

- Orders
- Inventory

Customer

- Profile
- Orders
- Wishlist
- Addresses

Features Deferred to Version 2

These are intentionally excluded from Version 1.

- Multi-vendor marketplace
- AI recommendations
- Live chat support
- Loyalty points
- Gift cards
- Multi-language
- Multi-currency
- Mobile app
- Product comparison
- Recently viewed products
- Subscription products

Keeping these out of Version 1 helps maintain a realistic project scope.

Final Technology Stack

Frontend

- React 19
- Vite
- Material UI
- React Router
- Axios
- React Hook Form
- Yup
- Context API

Backend

- Node.js
- Express.js
- JWT
- bcrypt
- Multer
- Cloudinary SDK
- Stripe SDK
- Nodemailer

Database

- MySQL
- Aiven MySQL

Deployment

- Frontend → Vercel
- Backend → Render
- Database → Aiven

Development Tools

- VS Code
- Git
- GitHub
- Postman
- MySQL Workbench
- PlantUML
- Mermaid
- Docker (later phase)
